

DROWSINESS DETECTION SYSTEM USING DEEP LEARNING

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Abstract-The National Highway Safety Administration estimates that lazy driving results in 1,550 passing, 71,000 wounds cause an excess of 100,000 mishaps every year. Teens as a group are at the epicenter of mishaps associated to sleepy driving; alongside other high-risk categories include patients with obstructive rest apnea (OSA) and people with similar problems, shift laborers, clinical house staff, policemen, and business drivers.

Keywords - Deep learning ,Python, Drowsiness Detection System, Tenserflow, CNN

I.INTRODUCTION

Drivers who are depleted of sleep in the driver's seat are answerable to around 40% of street mishaps, as revealed by a report from the Central Road Research Institute (CRRI) on the 300-km Agra-Lucknow Expressway. Human beings have always constructed machines and cooked ways to ease and cover their lives to rid them off of mundane conditioning like traveling to work or for more intriguing purposes like aircraft trips. As technology progressed, modes of transportation also progressed, and our reliance on these machines grew enormously. It has had a significant impact on our lives as we know it.

We are now able to go to destinations at a speed that our forebears could never have imagined. In today's society, practically everyone uses some form of transportation on a day to day basis. People who are wealthy enough possess cars, while others rely on public commutation facilities. Regardless of the socioeconomic position, there are certain set standards and canons of behavior for individuals who drive—such as staying attentive and active while driving. Neglecting these basic guidelines to ensure safer travel has resulted in a devastatingly huge amount of tragedies that are attributed to this unhealthy use of technology on a regular basis. Accurately obeying the rules, laws and regulations of the road may appear to most people as a minor detail, but it is a crucial one.

While on road, a technology has the most astounding power, and when in the wrong hands, it may be devastating; negligence can sometimes endanger the lives of others on the road. One such a kind of carelessness is failing to confess when we are too exhausted to drive.

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Several experiments have produced various kinds of research articles on driver drowsiness detection systems to help assist people from falling prey to such harmful outcome that stem from negligent recklessness.

However, some of the references and results generated by the system aren't always accurate. Consequently, this design has been used to provide more accurate data and also a different viewpoint on the problem at hand; to improve their executions and also to better optimize the result.

II. PROPOSED ALGORITHM

There are a wide range of products in the market that detect the driver's level of fatigue and are even implemented in various cars. This driver sleepiness detection system provides a similar functionality, but with more significant results and extra benefits. It also notifies the user when the sleepiness measure reaches beyond a saturation point.

Fatigue is a safety issue that has yet to be thoroughly investigated by many countries all over the globe despite its varying consequences. Fatigue, in general, is complicated to detect or observe, as opposed to alcohol and medication, for which they have readily available critical indicators and tests.

The foreseeable outcome of the problem of fatigue-related accidents should be increased awareness of tiredness-related accidents and also an encouragement for motorists to acknowledge their exhaustion when required. The former is complex and of high value, while the latter is impossible to accomplish without the former, as driving for an extended period of time is highly cost-efficient but risky.

Flow Diagram

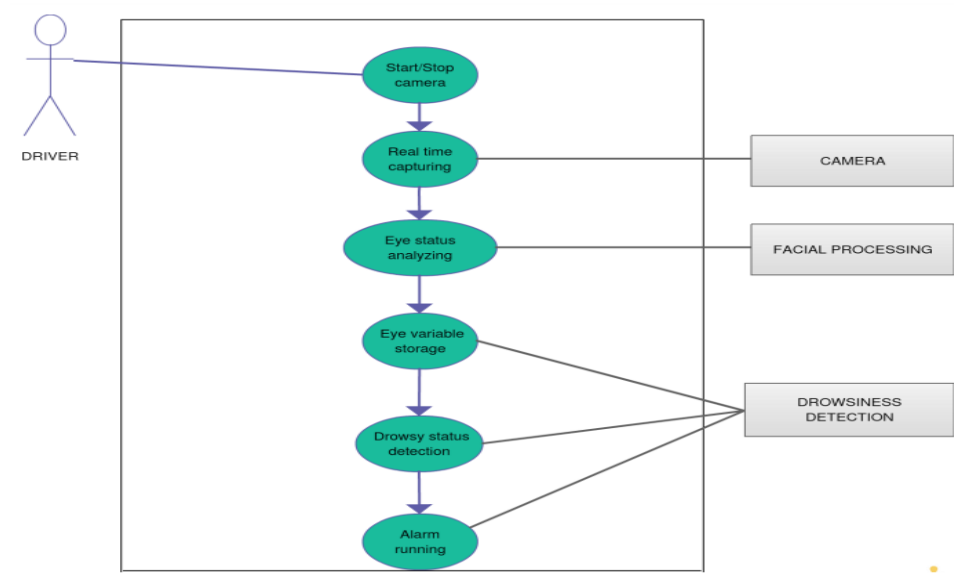


Figure 1 Flow Diagram

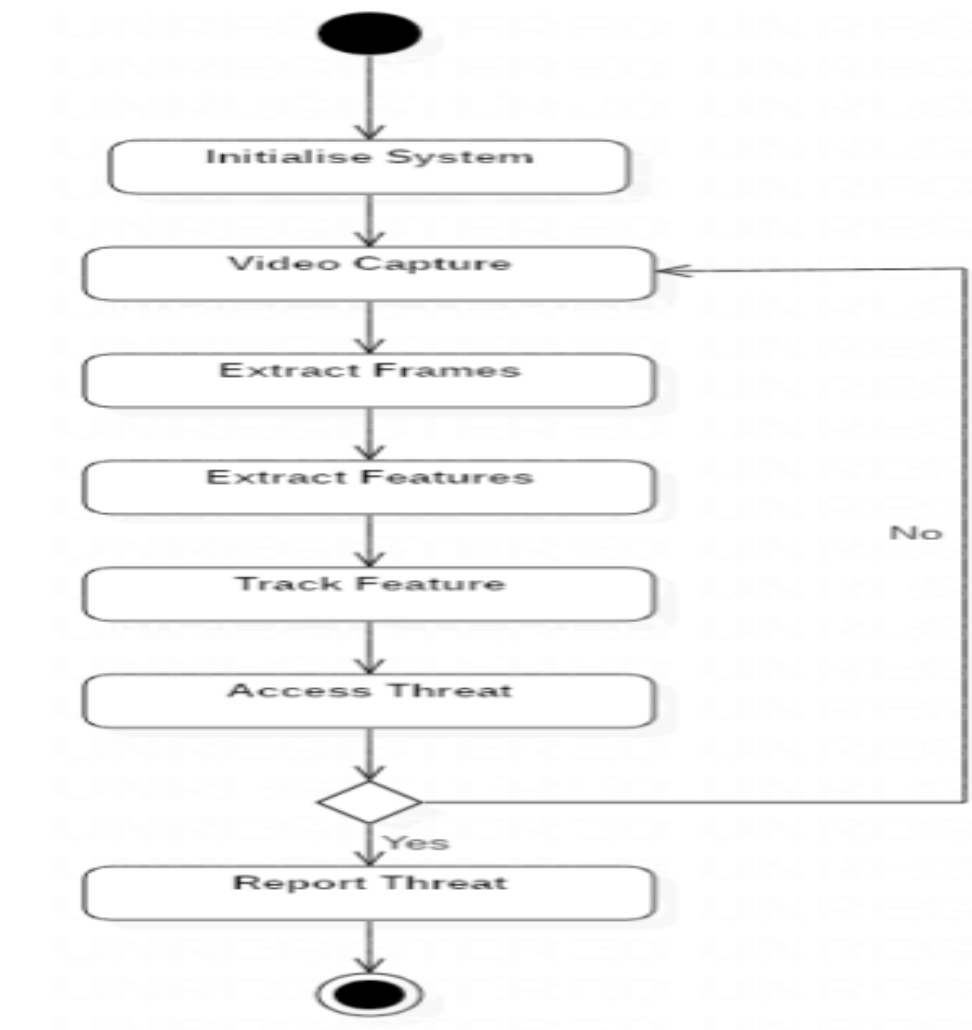
ACTIVITY DIAGRAM

Figure 2 Activity Diagram

SYSTEM MODEL

- The framework is created by making use of the incremental model.
- The center model of the framework is first created and afterwards augmented in this way at the time of testing with each turn.
- The underlying undertaking skeleton is then refined into expanding levels of ability
- At the subsequent incremental level, it might incorporate a new execution backing and improvements

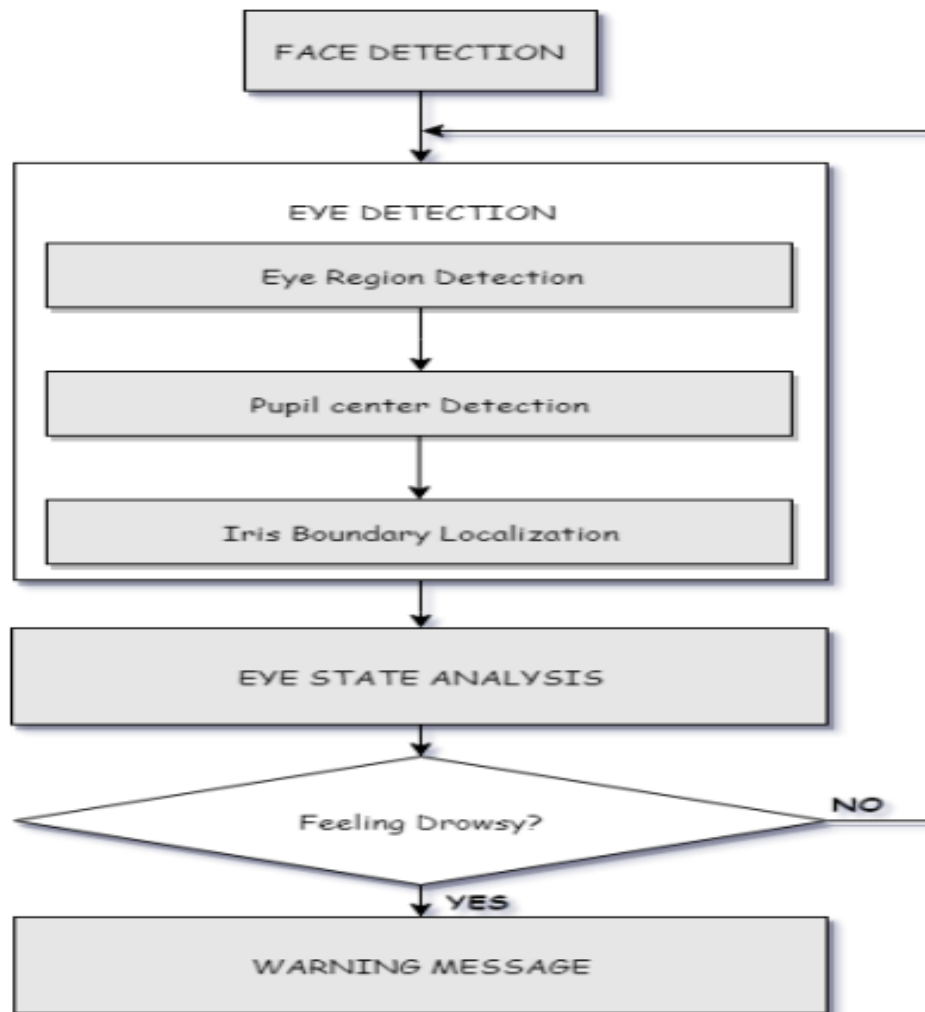


Figure 3 System Model

III.EXPERTMENT AND RESULTS

Creating Training Dataset and Testing Dataset

Step 1 – Collect images of eyes

- Collect images of eyes(Closed Eyes and Open Eyes) from different people (37 persons)
- Create separate folders for each individual.
- Each individual's images are saved in their respective folders.

Step 2– Renaming each image according to the formate

- Rename each image according to the formate
 # subject ID: xxx
 image number: xxx
 gender: 0 – male 1 – female
 glasses: 0 – no 1 – yes
 eye state: 0 – close 1 – open
 reflections: 0 – none 1 – low 2 – high

lighting conditions/image quality: 0 – bad 1 – good

sensor type: 01 – RealSense SR300 640x480 02 – IDS Imaging, 1280x1024 03 – Aptina
Imagin 752x480

example: s036_00123_0_1_0_1_1_02.png

Step 3 – Creating Training Dataset and Testing Dataset

- Create 2 folders

First folder contains images of open eyes

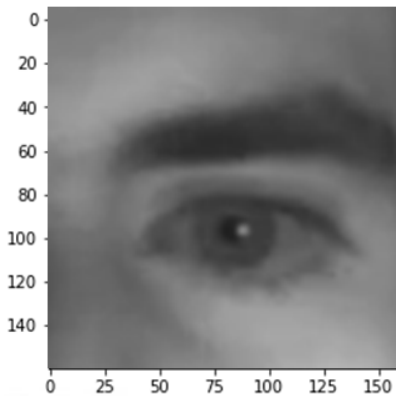


Figure 4 open eye

second folder contains images of closed eyes

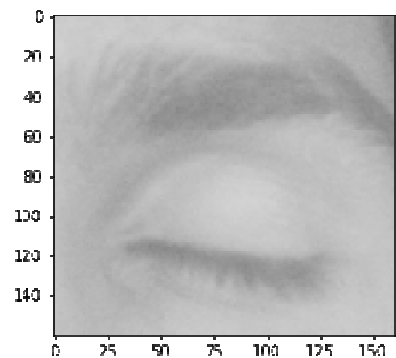


Figure 5 closed eye

- Move all the images of the open eyes and the closed eyes to their respective folders (Implementation of Algorithm)
- Then create two folders named “Train” and “Test” both these folders having two folders inside them (“open” and “closed”)
- Now move 10% of images to the test folder and 90% of images to the train folder, as shown in the below diagram.

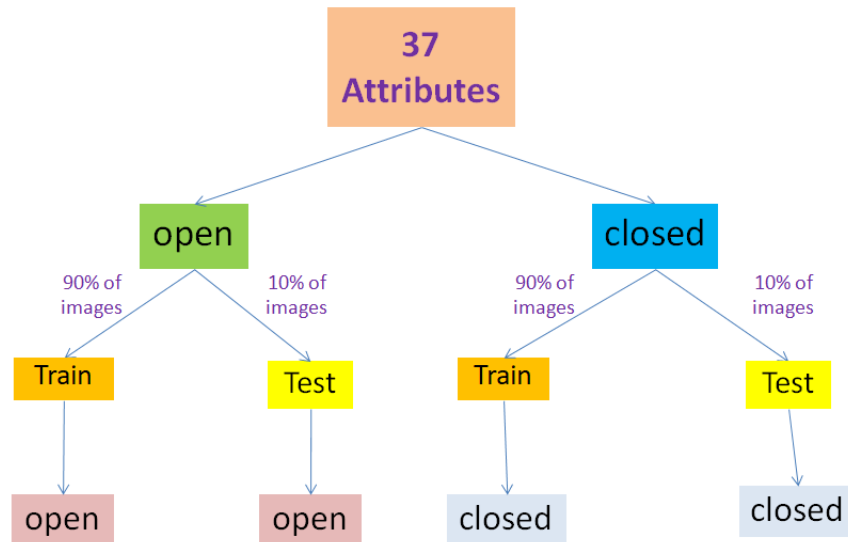


Figure 6 Creating Training Dataset and Testing Dataset

Real time implementation

Step 1 – Take a picture from a camera as an input.

- We'll use a camera and capture the photographs as an input.
- OpenCV, `cv2.VideoCapture(0)`
- Each frame is read with `cap.read()`, and the picture is saved in a frame variable.

Step 2 – Create a Region of Interest around the face in the photo (ROI).

- Make a grayscale conversion of the picture.
- Face detection with the haar cascade classifier
- `Face = cv2.CascadeClassifier('path to our haar cascade xml file')` is used to set our classifier.
- Detection using `faces = face.detectMultiScale(gray)`
- It produces an array of detections with x,y coordinates as well as height, which is the width of the object's border box.
- We can now go through the faces one by one and create boundary boxes for each one.

Step 3 – Feed the ROI to the classifier to detect the eye.

- In both `leye` and `reye`, set the cascade classifier for eyes.
Then using `left_eye = leye.detectMultiScale(gray)` detect the eyes.
- From the complete picture, we must now extract only the eyes data.
- The eye picture may then be extracted from the frame with this code. `l_eye` only contains the image data of the eye.
- This will be sent into our CNN classifier, which will determine whether or not the eyes are open. Similarly, the right eye will be extracted into `r_eye`.

Step 4 – If the eyes are open or closed, the classifier will classify them.

- The colour picture is converted to grayscale using `r_eye = cv2.cvtColor(r_eye, cv2.COLOR_BGR2GRAY)`.

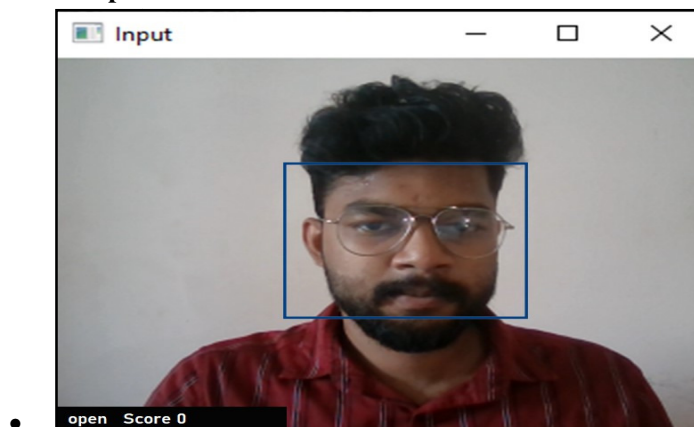
- We use `cv2.resize(r_eye, (24,24))` to resize the picture to 24*24 pixels because our model was trained on 24*24 pixel images.
- For improved convergence, we normalise our data ($r_eye = r_eye/255$) (All values will be between 0-1).
- To feed our classifier, expand the dimensions. We used `model = load_model('models/cnnCat2.h5')` to load our model.
- Now we use `model_lpred = model.predict_classes(l_eye)` to predict each eye.
- If the value of `lpred[0] = 1`, the eyes are open; if the value of `lpred[0] = 0`, the eyes are closed.

Step 5 – Calculate a score to see if the person is sleepy.

- Score -determine how long the person has closed his eyes
- We're using the `cv2.putText()` function to render the result on the screen, which will show the person's current condition in real time.
- A threshold is established, for example, if the score exceeds 15, it indicates that the person's eyelids have been closed for an extended amount of time. This is when we use sound to beep the sound. `play()`

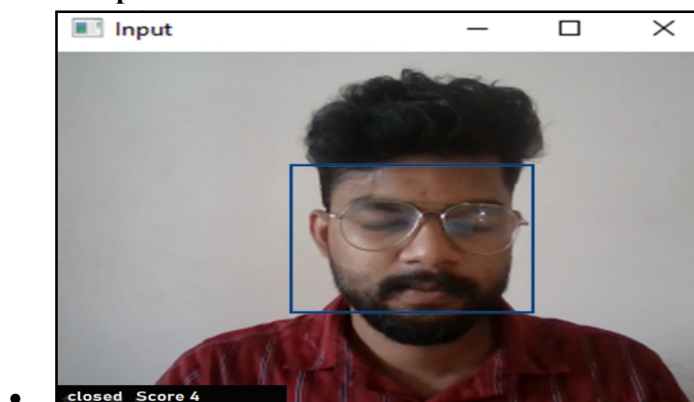
RESULTS

- **Eyes open – with spectacle**



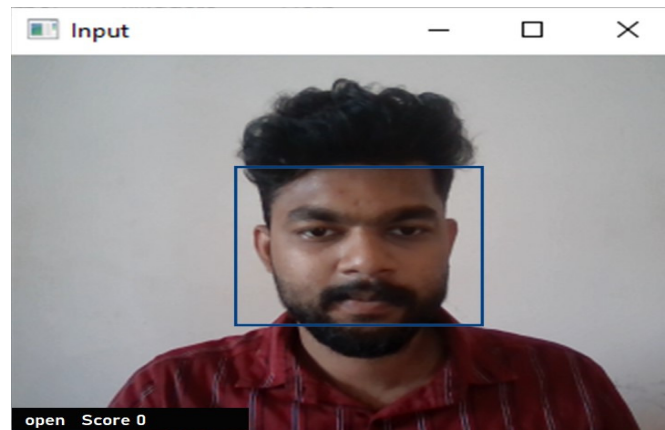
• **Figure 7 Eyes open – with spectacle**

- **Eyes closed – with spectacle**



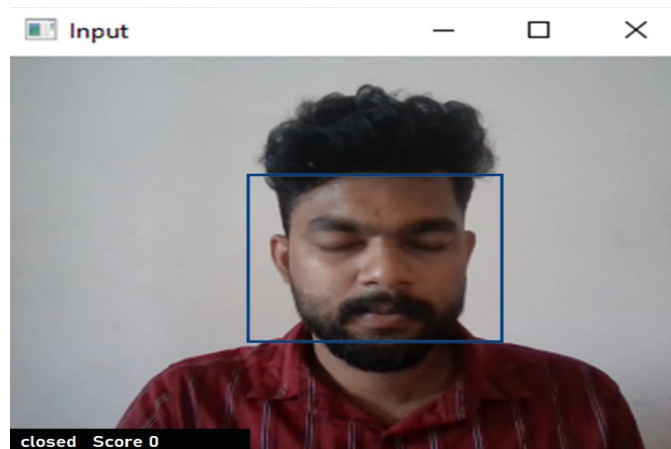
• **Figure 8 Eyes closed – with spectacle**

- Eyes open – without spectacles



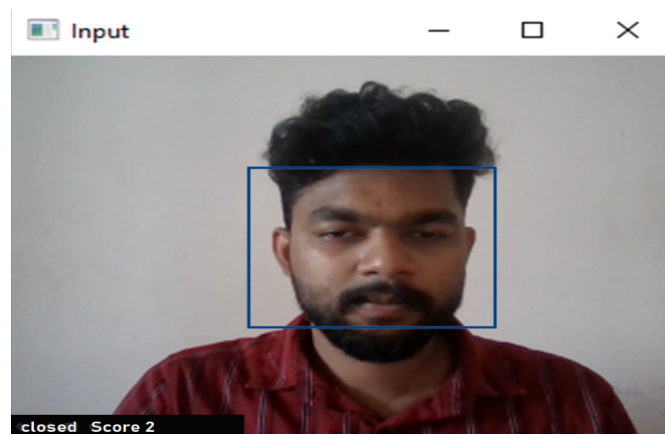
- Figure 9 Eyes open – without spectacle

- Eyes closed – without spectacle



- Figure 10 Eyes closed – without spectacle

- Eyes partially closed – without spectacle



- Figure 11 Eyes partially closed – without spectacle

IV.CONCLUSION

The system's objectives and criteria are being perfectly satisfied. The frame has reached a stable condition where all bugs have been eliminated. The frame-aware visitors who are familiar with the frame understand its main areas and, the fact that it alleviates tension for those suffering from fatigue-related disorders by informing them of their dozing posture while driving.

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